

SHIRISH MAN SHAKYA

DATA ANALYST

Sydney, NSW | 0492 877 385 | shirishshakya4@gmail.com
linkedin.com/in/shirish-man-shakya | github.com/Shakya658 | shakya658.github.io/portfolio | ACS Member

PROFESSIONAL SUMMARY

Data Analyst and recent graduate of the UTS Master of Data Science and Innovation (WAM 79, Academic Excellence Scholarship), with experience automating data extraction at Invisible Technologies and building SQL, Power BI and Python solutions around revenue, retention and customer behaviour. Built HarbourMetrics, a SaaS analytics platform covering MRR and churn, and developed a telecom churn solution that modelled a potential \$120,000 annual retention opportunity. Strong in SQL, Python and Power BI, with a track record of translating complex data into clear business insights and practical recommendations.

CORE SKILLS

Analytics and Business Intelligence: Power BI, DAX, Power Query, semantic modelling, dashboard development, KPI reporting, Tableau, Streamlit, Microsoft Excel

SQL and Data Analysis: SQL, PostgreSQL, complex queries, CTEs, window functions, cohort retention analysis, RFM segmentation, MRR and churn analysis

Python and Modelling: Python, Pandas, NumPy, Scikit-learn, XGBoost, SHAP, Optuna, feature engineering, predictive modelling, automated smoke testing

Statistics: Statistical analysis, hypothesis testing, statistical significance, regression, model evaluation and threshold optimisation

Data Engineering and Platforms: ETL and ELT pipelines, Medallion Architecture, Snowflake, GCP, Git and Linux

Data Quality and Communication: Data validation, data governance, requirements interpretation, process documentation, stakeholder communication and data storytelling

EDUCATION

Master of Data Science and Innovation | University of Technology Sydney | Sydney, NSW Jun 2023 - Jun 2025

- GPA 6.09/7.0 | WAM 79 | Postgraduate Academic Excellence Scholarship.
- High Distinctions in Data Science Practice, Data Visualisation and Narratives, Big Data Engineering, Deep Learning and Capstone Project.

Professional Year Program, Information Technology | QIBA | Sydney, NSW Jan 2026 - Present

- Completing a 12-month Australian industry-readiness program covering workplace communication, professional practice, career development and an employer internship.

Bachelor of Engineering, Computer Science | Nitte Meenakshi Institute of Technology | India Jun 2017 - Jun 2021

SELECTED PROJECTS

HarbourMetrics - SaaS Revenue and Retention Analytics Platform | PostgreSQL, Power BI, DAX, Semantic Modelling | [GitHub](#)

- Built a single view of revenue and retention performance for a simulated SaaS business with 2,000 customers across 24 months, enabling stakeholders to monitor MRR, subscriber growth and churn through an executive Power BI dashboard.
- Designed the PostgreSQL data model and analytical SQL layer using CTEs and window functions, developed DAX measures for core SaaS metrics, and identified an Enterprise-tier revenue concentration risk and a February Net MRR contraction.

Retail Sales Analytics Pipeline | PostgreSQL, SQL, Power BI, DAX, Medallion Architecture | [GitHub](#)

- Built a reusable analytics pipeline processing more than 1.6 million e-commerce transactions across nine related tables into Bronze, Silver and Gold data layers, with a five-page Power BI dashboard covering revenue, products, customers and operations.
- Identified that Champions represented 12.3% of customers but generated 21.1% of revenue, while Potential Loyalists accounted for 24.8% of the customer base but only 8.6% of revenue, and flagged Victoria as a logistics underperformer with an 86.82% on-time delivery rate.

Customer Churn Prediction and Revenue Recovery | Python, XGBoost, Scikit-learn, SHAP, Streamlit | [GitHub](#)

- Analysed churn risk across 7,043 telecom customer records and modelled a potential \$120,000 annual retention opportunity under targeted intervention assumptions.
- Compared XGBoost, Random Forest and Logistic Regression using cross-validation and more than 15 engineered behavioural features, achieving ROC AUC 0.85 and 81% accuracy.
- Deployed a Streamlit application for customer-level scoring, model comparison, threshold-based classification and SHAP explanations, supported by automated model-artefact tests.

30-Day Hospital Readmission Risk Model | Python, XGBoost, SHAP, Optuna, Streamlit | [GitHub](#)

- Processed 101,766 raw hospital encounters and engineered a 95-feature modelling dataset, applying Optuna tuning and stratified cross-validation to achieve ROC AUC 0.665 on a highly imbalanced classification problem.
- Deployed a SHAP-powered Streamlit interface that translated patient-level predictions into clear risk explanations for non-technical users.
- Added automated tests validating saved model artefacts, feature ordering, scaler compatibility and probability outputs.

Cancer Risk Stratification via Multi-Omics | Python, PCA, Clustering, SHAP | UTS Capstone

- Awarded a High Distinction for identifying three distinct patient risk groups from multi-omics data covering 1,200 colorectal cancer patients.
- Reduced the feature space by 80% using PCA while retaining meaningful clinical information and used SHAP analysis to identify influential biomarkers across patient groups.

PROFESSIONAL EXPERIENCE

Data Analyst | Invisible Technologies | Remote, Contract Aug 2022 - Jul 2024

- Automated the extraction and structuring of billing data from PDF documents using Python and pdfplumber, reducing reliance on manual data entry and accelerating the preparation of analysis-ready outputs.
- Validated extracted records for accuracy, completeness and consistency before they entered client reporting and compliance workflows.
- Standardised data-processing workflows across enterprise client engagements, reducing manual handling and improving turnaround time.

IT Systems and Data Supervisor | Tripura Secondary School | Kathmandu, Nepal Mar 2020 - Aug 2022

- Designed and maintained school databases, reporting processes and IT systems, and led the migration to remote operations during COVID-19 for more than 50 students and staff.

Night Fill Team Member | Coles Supermarkets | Sydney, NSW Jul 2023 - Present

- Maintained approximately 20 hours of paid employment per week while completing a full-time master's degree, demonstrating reliability, prioritisation and sustained performance under competing commitments.

SELECTED PROFESSIONAL DEVELOPMENT

- Microsoft PL-300 Power BI Data Analyst: Exam preparation in progress
- Advanced Python for Data Analysis | LinkedIn Learning
- Building Data Visualisations | LinkedIn Learning
- Machine Learning with Python: Foundations | LinkedIn Learning

ADDITIONAL INFORMATION

Work rights: Full Australian working rights under a Temporary Graduate visa (subclass 485)

Languages: English - PTE Academic Overall 90 | Nepali - Native